

Advanced (Carolina Reaper)**Q1.** Evaluate the expression $3 \cdot 2 + 12 \div 4$.

- (A) 9
(B) 10
(C) 11
(D) 12
(E) 13

(A) 20.75

- (B) 21.25
(C) 22.50
(D) 23.75
(E) 24.25

Q2. Translate the phrase '5 more than 3 times a number' into an algebraic expression.

- (A) $3x + 5$
(B) $3x - 5$
(C) $5x + 3$
(D) $5x - 3$
(E) $x + 15$

Q6. A person wants to buy a new bike that costs 120. *If they have a* 20 coupon and 10

- (A) 100
(B) 104
(C) 106
(D) 108
(E) 110

Q3. Simplify the expression $2x + 5x - 3x$.

- (A) $4x$
(B) $6x$
(C) $8x$
(D) $10x$
(E) $12x$

Q7. Evaluate the expression $(2x + 3) + (5x - 2)$.

- (A) $7x + 1$
(B) $7x - 1$
(C) $6x + 1$
(D) $6x - 1$
(E) $5x + 1$

Q4. Evaluate the expression $x^2 + 4x - 5$ when $x = 2$.

- (A) 3
(B) 5
(C) 7
(D) 9
(E) 11

Q8. A coffee shop has a promotion where customers can buy a large coffee for 2.50 and get a small coffee for 1.50. If a customer buys 2 large coffees and 2 small coffees, how much will they pay in total?

- (A) 8
(B) 9
(C) 10
(D) 11
(E) 12

Q5. A bookstore is having a sale where everything is 15**Open Ended Questions (FRQ)****Q9.** A person is planning a road trip that is 240 miles long. If they drive at an average speed of 40 miles per hour, how many hours will the trip take? Use the expression $time = \frac{distance}{speed}$ to solve the problem. Show all work and calculations.**Q10.** A water tank can hold 1200 gallons of water. If 300 gallons of water are already in the tank, and 50 gallons of water are added every hour, how many hours will it take to fill the tank? Use the expression $total\ water = initial\ water + (rate \cdot time)$ to solve the problem. Show all work and calculations.**Chef's Tips**

- Make sure students have a calculator
- Encourage students to read the questions carefully
- Remind students to show all work and calculations for FRQs